

Electrical Characterization and Study of Radiation Effects on High Speed Comparator

Submitted By

**VIVEK A (Enroll No. MGP24EVLSI026)
B.Tech in Electronics Engineering (VLSI Design and Technology)**

Under the Supervision of

**MR. SHUBHAM CHAUHAN
(SCIENTIST / ENGINEER 'SD')
VLSI Test and Application Development Division**



**Semi-Conductor Laboratory
Ministry of Electronics and Information Technology (MeitY)
Government of India**



Saintgits College of Engineering, Kottayam, Kerala

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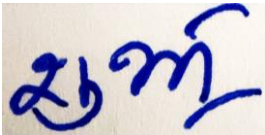
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VIVEK A
MGP24EVLSI026
Saintgits College of Engineering, Kottayam, Kerala

Dated:02/07/2026

Certified that the above statement made by the Trainee is correct to the best of my knowledge and belief.



Shubham Chauhan
Scientist/ Engineer 'SD'
VLSI Test and Application Development Division (VTAD)
Semi-Conductor Laboratory, S.A.S Nagar
Government of India

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I express my sincere gratitude to **Semi-Conductor Laboratory (SCL), Mohali**, for providing me with the opportunity to undertake my summer internship in the **VLSI Test & Application Development Division**. The internship offered valuable exposure to semiconductor device characterization, automated testing techniques, and radiation reliability assessment, enabling me to bridge the gap between theoretical knowledge and industrial practice.

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Abstract

High-speed comparators are fundamental building blocks in modern analog and mixed-signal integrated circuits, where rapid and accurate decision-making is essential for high-performance electronic systems. Their applications span data converters, communication systems, instrumentation, automotive electronics, medical devices, and aerospace technologies. As operating frequencies continue to increase, comprehensive electrical characterization and reliability assessment become critical to ensure dependable device performance under varying operating conditions.

The primary objective of this internship was to characterize an SCL-designed high-speed comparator and study the methodology adopted for radiation testing of high-speed comparator integrated circuits. The characterization process involved evaluating important static and dynamic performance parameters, including positive and negative trip voltages V_{Trip+} and V_{Trip-} , input offset voltage, common-mode rejection ratio (CMRR), power supply rejection ratio (PSRR), propagation delay, rise time, fall time, and other timing characteristics.

The experimental setup consisted of a digital oscilloscope, arbitrary function generator, source measure unit, digital multimeter, and relay switching hardware controlled through an Arduino interface. Instrument communication was established using the Virtual Instrument Software Architecture (VISA) protocol, enabling automated measurement and data acquisition through MATLAB. Instead of manual measurements, automation improved repeatability, reduced characterization time, and minimized operator-induced errors.

In addition to electrical characterization, the internship included a study of radiation testing methodologies applicable to space-qualified semiconductor devices. Particular emphasis was placed on Biased High Dose Rate (HDR) Total Ionizing Dose testing performed according to MIL-STD-883 Test Method 1019.9 Condition A. The study provided insight into radiation-induced degradation mechanisms, test procedures, bias conditions, and post-irradiation electrical evaluation of comparator devices intended for harsh radiation environments.

The internship provided comprehensive exposure to semiconductor device testing, automated instrumentation, analog integrated circuit characterization, and reliability evaluation. The knowledge and practical skills acquired during this work significantly enhanced understanding of industrial characterization methodologies employed for high-speed analog integrated circuits.

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Chapter 1 : Company Profile

1.1 Overview:

Semi-Conductor Laboratory (SCL), an autonomous body under the Ministry of Electronics and Information Technology (MeitY), Government of India, is the only Integrated Device Manufacturing Facility in the country providing end-to-end solutions for Development of Application Specific Integrated Circuits (ASICs), Opto-electronics devices, and Micro Electro Mechanical System (MEMS) Devices encompassing Design, Fabrication, Assembly, Packaging, Testing, and Reliability Assurance. Formerly known as Semiconductor Complex Limited, a Government of India enterprise, it was converted into Semi-Conductor Laboratory under the Department of Space, Government of India, w.e.f. September 1, 2006. Administrative control of SCL, Society was transferred from the Department of Space (DoS) to the Ministry of Electronics and Information Technology (MeitY) vide notification No. 1/21/1/2022-Cab dated February 07, 2022.



Figure 1.1: Semi-Conductor Laboratory Main Gate

SCL has an 8" wafer fabrication line qualified to the JEDEC-JP001A standard with a 180 nm CMOS technology node. SCL also has a 6" fabrication line for MEMS development and is expanding it to include a compound semiconductor fabrication facility. Process capability at SCL enables a 1.8V, 1.8/3.3V, or 1.8V/5V power-supply solution with 4-6 Al-metal layers and analog modules. The VLSI design domain in SCL spreads over analog, digital, mixed-signal, memory, RF-CMOS, and optoelectronic in the form of silicon-proven and space-qualified ASICs, ASSPs, SoCs, SCL excels in developing ceramic packages and meets the demanding test requirements at the wafer and package level, along with test plan development for high-pin-count integrated circuits, RF, and MEMS devices. At SCL, quality and reliability assurance adhere to global performance specifications such as MIL-PRF-38535, JEDEC-JP001A, and MIL-STD-883. SCL possesses capabilities in power, water, and air management, bulk, and specialty gas

distribution systems. The quality parameters of Ultra Pure Water (UPW) and bulk gases produced at SCL are at par with international standards. SCL brings decades of experience to provide customers with unparalleled microelectronics solutions in India. SCL is also engaged in the fabrication of Hi-Reliability boards, Radio systems, and the indigenization of electronic subsystems.

The activities carried out at SCL encompass nearly every stage of semiconductor development. These include:

1. Integrated Circuit Design

The design teams at SCL develop analog, digital, RF, and mixed-signal integrated circuits for various industrial and strategic applications. The design process includes circuit simulation, layout generation, verification, and design validation before fabrication.

2. Wafer Fabrication

SCL possesses advanced semiconductor fabrication facilities capable of manufacturing CMOS-based integrated circuits using sophisticated lithography, oxidation, diffusion, ion implantation, metallization, and passivation processes. Strict cleanroom protocols are maintained to ensure high manufacturing yield and device reliability.

3. Packaging

After fabrication, semiconductor dies are assembled into suitable packages using wire bonding, encapsulation, and hermetic sealing techniques depending on the intended application.

4. Electrical Characterization

Each fabricated device undergoes extensive electrical testing to verify compliance with design specifications. Static, dynamic, and timing characteristics are measured using precision automated instrumentation.

5. Reliability Testing

Reliability evaluation includes environmental testing, burn-in, life testing, thermal cycling, radiation testing, and electrical stress analysis to ensure device performance under harsh operating conditions.

6. Failure Analysis

Devices that fail qualification tests undergo microscopic and electrical failure analysis to identify manufacturing defects or design-related issues. These findings are used to improve future fabrication processes.

1.2 VLSI Test & Application Development Division:

The internship was conducted in the **VLSI Test & Application Development Division (VTAD)**, which is responsible for evaluating the functional and electrical performance of integrated circuits fabricated at SCL.

The division performs activities such as:

- Development of electrical test methodologies.
- Automated device characterization.
- Wafer-level testing.
- Package-level testing.
- Performance validation.
- Analog and mixed-signal IC characterization.
- Timing parameter evaluation.
- Reliability qualification.
- Radiation test preparation.
- Automated test software development.

One of the important responsibilities of the division is to ensure that fabricated ICs meet the required electrical specifications before being qualified for practical applications.



Figure 1.2 VTAD LAB

1.3 Reason for Interning at this Organization:

1. Exposure to Advanced Semiconductor Technology

SCL is one of India's leading semiconductor fabrication facilities, providing valuable exposure to advanced chip design, fabrication, and testing processes that are not typically accessible in academic environments.

2. Practical Application of Theoretical Knowledge

The internship offers an excellent opportunity to apply concepts learned in electronics, VLSI design, and microelectronics to real-world industrial systems, helping bridge the gap between theory and practice.

3. Access to State-of-the-Art Infrastructure

SCL is equipped with modern clean room facilities and sophisticated fabrication equipment, allowing interns to observe and understand high-precision manufacturing environments used in semiconductor production.

4. Learning from Experienced Professionals

Working alongside skilled scientists/engineers provides valuable mentorship, practical insights, and exposure to industry best practices beyond classroom learning.

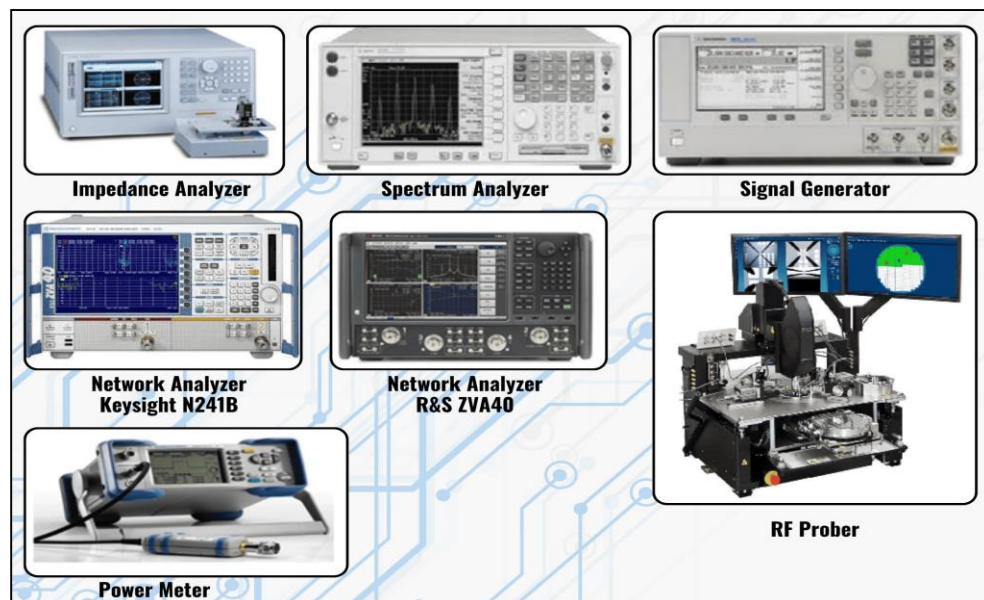


Figure 1.3 RF Test Facilities

5. Enhancement of Technical and Analytical Skills

Through hands-on exposure and observation, interns develop problem-solving abilities, analytical thinking, and a deeper understanding of device behaviour and fabrication challenges.



Figure 1.4 RF Test Facilities

6. Career Development and Industry Readiness

The experience helps in building professional discipline, improving technical competence, and preparing for future roles in the semiconductor and VLSI industries.

7. Contribution to India's Growing Semiconductor Ecosystem

With increasing focus on domestic semiconductor manufacturing, this internship provides an opportunity to be part of a rapidly growing industry and understand its future potential and challenges.

1.4 Objectives of the Internship:

The primary objectives of the internship were:

- To understand the operating principles of high-speed comparator circuits.
- To study the architecture and functional behaviour of an SCL-designed high-speed comparator.
- To perform electrical characterization using modern laboratory instruments.
- To automate the characterization procedure using MATLAB and VISA-based instrument communication.
- To evaluate important static and dynamic electrical parameters of the comparator.
- To understand laboratory automation techniques used in semiconductor industries.
- To study radiation effects on semiconductor integrated circuits.
- To understand Biased High Dose Rate radiation testing according to MIL-STD-883 Test Method 1019.9 Condition A.

- To compare measured device performance with expected specifications.

1.5 Importance of Device Characterization:

Device characterization is a critical step in semiconductor product development. Even after successful fabrication, every integrated circuit must be evaluated experimentally to verify that its electrical performance matches the intended design specifications.

Characterization provides information regarding:

- Switching performance
- Offset voltage
- Noise behaviour
- Timing characteristics
- Temperature dependence
- Supply voltage sensitivity
- Process variation
- Reliability under operating conditions

The information obtained through characterization helps designers improve future versions of the integrated circuit while ensuring compliance with customer and industrial requirements.

For high-speed comparators, characterization becomes particularly important because very small variations in propagation delay or input offset voltage can significantly affect system performance in applications such as analog-to-digital converters, communication systems, and high-speed signal processing.

1.6 Learning Outcomes:

The internship provided valuable exposure to both theoretical concepts and industrial practices. The major learning outcomes include:

- Fundamentals of high-speed comparator operation.
- Semiconductor device characterization techniques.

- Practical use of digital oscilloscopes, arbitrary function generators, source measure units, and digital multimeters.
- Automation of laboratory measurements using computer-controlled instrumentation.
- VISA-based communication between instruments and a host computer.
- Relay switching and test setup configuration.
- Measurement of comparator performance parameters.
- Timing analysis using digital oscilloscopes.
- Radiation reliability concepts applicable to space electronics.

Industrial documentation and reporting practices

Chapter 2 : Introduction to High Speed Comparators

2.1 Introduction

A comparator is a high-speed analog circuit that compares two input voltages and generates a digital output indicating which input has the higher potential. Unlike an operational amplifier operating in the linear region, a comparator is designed to switch rapidly between logic levels, making it suitable for high-speed signal processing applications.

High-speed comparators are widely used in Analog-to-Digital Converters (ADCs), communication systems, sensor interfaces, memory circuits, and clock recovery applications, where rapid and accurate voltage comparison is essential. Modern integrated circuits employ latched comparators because they offer high operating speed, low power consumption, and improved immunity to noise.

The comparator characterized during this internship was a **high-speed latched comparator designed and fabricated at Semiconductor Laboratory (SCL), Mohali**. The objective of this chapter is to present the basic operating principles required to understand the electrical characterization methodology discussed in the subsequent chapters.

2.2 Operating Principle of a High-Speed Latched Comparator

A comparator compares the voltages applied to its two input terminals and produces a binary output based on their relative magnitudes. If the voltage at the non-inverting input exceeds the voltage at the inverting input, the output switches to a logic HIGH state. Conversely, if the inverting input voltage is greater, the output transitions to a logic LOW state.

In high-speed applications, continuous comparison is often replaced by **clock-controlled (latched) operation**. A latch-enable signal determines the instant at which the comparator samples the input voltages. During the evaluation phase, the comparator rapidly amplifies the small differential input using positive feedback and latches the result at the output. This regenerative action enables extremely fast decision-making while minimizing power consumption.

Compared to conventional comparators, latched comparators provide:

- High operating speed
- Low power dissipation

- Reduced offset due to regenerative amplification
- Suitability for high-frequency applications
- Improved immunity to switching noise

These characteristics make them suitable for high-speed integrated circuits where timing accuracy is critical.

2.2 Comparator Fundamentals

A comparator is an electronic circuit that compares two input voltages and determines which one is larger. Based on this comparison, the output switches between two distinct logic levels representing either a HIGH or LOW state.

The comparator has two input terminals:

- Non-inverting input (+)
- Inverting input (−)

The output is governed by the relationship between these two input voltages.

If

$$V_+ > V_-$$

the output switches to the HIGH logic level.

If

$$V_+ < V_-$$

the output switches to the LOW logic level.

Unlike operational amplifiers, comparators are specifically optimized for high-speed switching rather than linear amplification. Their internal circuitry is designed to minimize decision time and maximize transition speed, making them suitable for applications operating at frequencies ranging from a few kilohertz to several gigahertz.

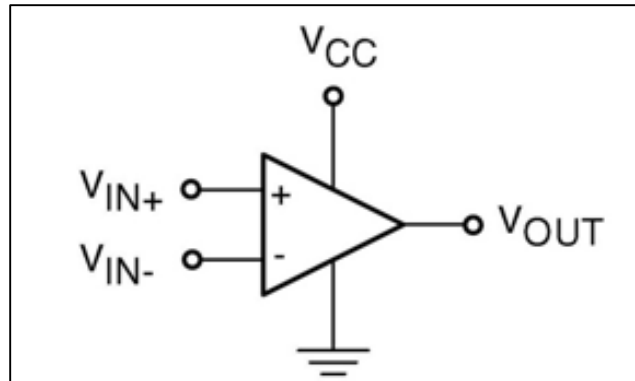


Figure 2.1 Basic Comparator Symbol

2.3 Principle of Operation

The operating principle of a comparator is based on differential amplification.

Initially, the comparator senses the voltage difference

$$V_d = V_+ - V_-$$

where

- V_+ = Non-inverting input voltage
- V_- = Inverting input voltage

This differential voltage is amplified by an internal high-gain differential amplifier. Since the voltage gain is extremely large, even microvolt-level differences are amplified sufficiently to force the output towards one of the supply rails.

Thus, the comparator behaves as a binary decision-making circuit.

Instead of reproducing the input waveform, it determines only which input is larger.

Because of this characteristic, comparators are often called **one-bit analog-to-digital converters**.

2.4 Comparator versus Operational Amplifier

Although operational amplifiers and comparators appear similar externally, they are designed for different purposes.

Comparator	Operational Amplifier
Open-loop operation	Closed-loop operation
Very high switching speed	Moderate speed
Digital output	Analog output
Optimized for fast transitions	Optimized for linear amplification
Minimal propagation delay	Frequency compensation included
Usually no compensation capacitor	Internally compensated

Table 2-1 Comparator vs Operational Amplifier

Operational amplifiers contain internal frequency compensation to ensure stable linear operation under negative feedback. This compensation limits their speed.

Comparators eliminate this compensation, allowing significantly faster switching and making them suitable for high-speed decision-making applications.

2.5 Applications of High-Speed Comparators

High-speed comparators play a crucial role in numerous electronic systems where rapid signal processing is required.

Some important applications include:

Analog-to-Digital Converters (ADCs)

Flash ADCs require a large number of comparators operating simultaneously. The overall conversion speed depends directly on comparator performance.

High-Speed Communication Systems

Comparators are employed in clock recovery circuits, data receivers, optical communication systems, and equalizers where signal transitions occur within nanoseconds.

Radar Systems

Pulse detection and timing circuits use comparators for identifying weak reflected signals with high precision.

Medical Electronics

Biomedical instruments utilize comparators for threshold detection in ECG, EEG, and imaging systems.

Automotive Electronics

Comparators are used in engine control units, battery management systems, collision avoidance sensors, and motor controllers.

Aerospace and Space Electronics

Radiation-hardened comparators are employed in satellites, launch vehicles, and spacecraft instrumentation because of their ability to operate reliably in harsh radiation environments.

Industrial Automation

High-speed control systems, motor drives, and protection circuits use comparators for rapid fault detection and real-time monitoring.

2.6 Characteristics of High-Speed Comparators

A high-speed comparator is designed to achieve extremely fast switching while maintaining high accuracy.

Some of its important characteristics include:

- Low propagation delay
- Low input offset voltage
- Small differential delay
- High slew rate
- Wide input common-mode range
- High CMRR
- High PSRR

- Low power consumption
- High noise immunity
- Stable operation over temperature

These characteristics determine the overall performance of the comparator in practical applications.

Static Parameters

Static parameters describe the behaviour of the comparator under steady-state or slowly varying input conditions. These include:

- Positive Trip Voltage (V_{Trip+})
- Negative Trip Voltage (V_{Trip-})
- Input Offset Voltage
- Common-Mode Rejection Ratio (CMRR)
- Power Supply Rejection Ratio (PSRR)

Dynamic Parameters

Dynamic parameters describe the response of the comparator to rapidly changing input signals. These include:

- Rise Time
- Fall Time
- Propagation Delay
- Differential Delay
- Setup Time
- Hold Time
- Pulse Width

Dynamic parameters are particularly important in high-speed communication and data-conversion systems, where timing accuracy directly impacts system performance.

2.7 Internal Functional Blocks of a High-Speed Comparator

Although different comparator architectures exist, a typical high-speed comparator consists of the following functional blocks:

1. Differential Input Stage
2. Gain Stage
3. Regenerative Latch
4. Output Buffer
5. Bias Circuit
6. Reference Circuit
7. Output Driver

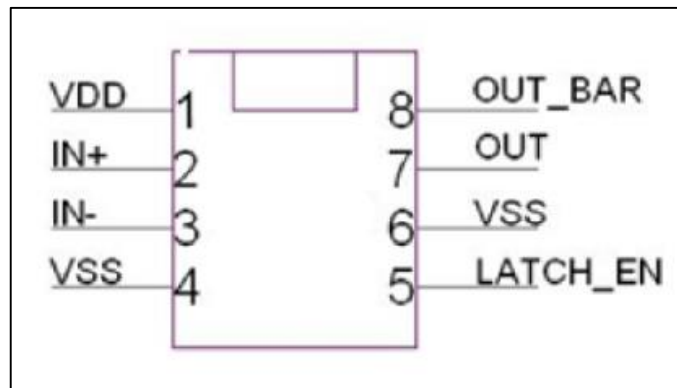


Figure 2.2 Internal Functional Block Diagram of a High-Speed Comparator

Differential Input Stage

The differential input stage receives the two input voltages and amplifies only their difference while rejecting common-mode signals. This stage largely determines the input offset voltage, input bias current, and common-mode rejection ratio of the comparator.

Gain Stage

The gain stage provides additional amplification to the differential signal. High gain ensures that even very small input differences produce a decisive output transition.

Regenerative Latch

The regenerative latch is one of the most critical blocks in high-speed comparators. It employs positive feedback to accelerate the decision-making process. Once the

differential input exceeds a small threshold, regenerative action rapidly drives the output to a stable logic state, reducing propagation delay and improving switching speed.

Output Buffer

The output buffer isolates the sensitive internal circuitry from external loads and provides sufficient drive capability for digital systems while preserving fast transitions.

Bias Circuit

The bias circuit generates stable internal bias currents and voltages that ensure consistent comparator performance across variations in temperature, supply voltage, and manufacturing process.

2.8 Latch Enable Operation

The comparator characterized during this internship incorporates a **Latch Enable (LE)** signal.

The latch enable controls when the comparator samples the input and updates its output.

During characterization:

- A **100 Hz triangular waveform** was applied to the comparator input.
- A **100 kHz square wave** was applied to the latch enable input.

When the latch enable becomes active, the comparator compares the instantaneous input voltage with the reference voltage and updates its output. During the inactive period, the output remains latched and retains its previous logic state.

This sampling operation is particularly useful in high-speed digital systems because it synchronizes comparator decisions with a system clock, improving timing accuracy and reducing noise sensitivity.

2.9 Importance of Comparator Characterization

Characterization is the process of experimentally evaluating the electrical performance of a fabricated integrated circuit under specified operating conditions.

For high-speed comparators, characterization verifies whether the fabricated device satisfies its design specifications.

The process includes measurement of:

- Positive trip voltage (V_{Trip+})
- Negative trip voltage (V_{Trip-})
- Input offset voltage
- Common-mode rejection ratio
- Power supply rejection ratio
- Rise time
- Fall time
- Propagation delay
- Differential delay
- Pulse width

Accurate characterization is essential because fabrication process variations, temperature changes, and supply fluctuations can significantly influence comparator performance.

2.10 Motivation for the Present Work

The increasing demand for high-speed analog integrated circuits necessitates reliable electrical characterization methods capable of accurately measuring static and dynamic device parameters. Manual characterization is often time-consuming, susceptible to human error, and difficult to reproduce consistently.

During this internship, an automated characterization setup was employed using laboratory instruments interfaced with a host computer. Automation enabled efficient acquisition of measurement data while improving repeatability and reducing overall testing time.

In addition to electrical characterization, the study also included an investigation of radiation testing procedures used for qualifying semiconductor devices intended for space applications. Understanding the effects of ionizing radiation on comparator performance is essential for ensuring reliable operation in harsh environments.

Chapter 3 : Testing Methodology

3.1 Introduction

The performance of a high-speed comparator is evaluated by measuring a number of electrical parameters that describe its static accuracy, dynamic response, noise immunity, and sensitivity to external disturbances. These parameters collectively determine whether the fabricated comparator satisfies its design objectives and can operate reliably under different environmental and electrical conditions.

During this internship, the SCL-designed high-speed comparator was characterized using an automated laboratory setup. A combination of a digital oscilloscope, arbitrary function generator (AFG), source measure unit (SMU), digital multimeter (DMM), and relay module was employed to measure the comparator's electrical characteristics. MATLAB was used to automate the measurements and acquire data from the instruments through VISA communication.

The important parameters evaluated during characterization are discussed in the following sections.

3.2 Positive Trip Voltage (V_{Trip+})

Definition

The **Positive Trip Voltage (V_{Trip+})** is defined as the input voltage at which the comparator output changes state while the input voltage is increasing.

When the input voltage gradually rises and crosses the internal switching threshold of the comparator, the output transitions from one logic state to the other. The corresponding input voltage is called the positive trip voltage.

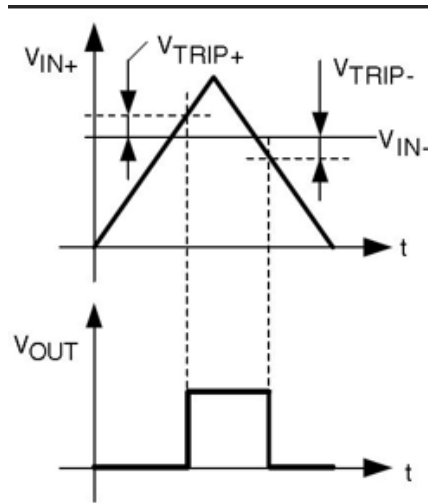


Figure 3.1 Determination of Positive Trip Voltage

Importance

Positive trip voltage determines the comparator's switching threshold during an increasing input signal. Accurate knowledge of this parameter is important because it directly affects threshold detection accuracy.

Applications requiring precise voltage comparison, such as Analog-to-Digital Converters (ADCs), rely heavily on accurate trip voltage.

Measurement Principle

A low-frequency triangular waveform is applied to the comparator input.

As the voltage increases slowly, the comparator output changes state at a particular input voltage.

The corresponding value of the input voltage is recorded as **V_{Trip+}**.

During the internship, this value was obtained automatically from the oscilloscope waveform using MATLAB-based processing.

Factors Affecting V_{Trip+}

- Input offset voltage
- Process variation

- Temperature
- Supply voltage
- Noise

3.3 Negative Trip Voltage (V_{TRIP-})

Definition

The **Negative Trip Voltage (V_{TRIP-})** is the input voltage at which the comparator changes state while the input voltage is decreasing.

Unlike V_{TRIP+} , this parameter is measured during the falling portion of the triangular input waveform.

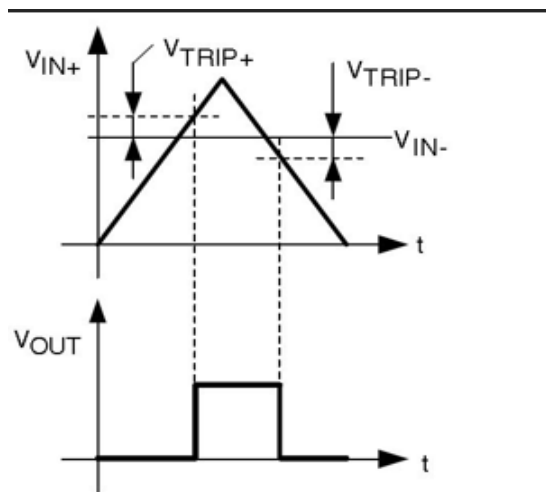


Figure 3.2 Determination of Negative Trip Voltage

Importance

V_{TRIP-} helps determine the switching threshold for decreasing input voltages.

The difference between V_{TRIP+} and V_{TRIP-} indicates the hysteresis of the comparator.

Measurement Method

A triangular waveform is continuously applied.

During the falling portion of the waveform, the comparator output switches.

The corresponding input voltage is recorded as V_{Trip-} .

3.4 Input Offset Voltage (VIO)

Definition

Ideally, a comparator should switch exactly when both input voltages are equal.

However, due to unavoidable transistor mismatches during fabrication, the comparator switches at a small non-zero differential voltage.

This voltage is known as the **Input Offset Voltage (VIO)**.

Mathematically,

$$VIO = (V_{TRIP+} + V_{TRIP-}) / 2$$

Significance

Input offset voltage represents the intrinsic error of the comparator.

Smaller offset voltage indicates higher comparator accuracy.

High-speed precision applications demand extremely low offset voltage.

Causes

- Device mismatch
- Threshold voltage variation
- Layout asymmetry
- Temperature variation

Measurement

The positive and negative trip voltages are measured first.

The input offset voltage is then calculated using the above equation.

Advantages

- Reduces output chatter
- Improves switching stability
- Increases immunity to electrical noise

3.5 Common Mode Rejection Ratio (CMRR)

Definition

The Common Mode Rejection Ratio represents the ability of the comparator to reject signals that appear simultaneously on both input terminals.

It is expressed as

$$CMRR = 20 \log \left(\frac{A_d}{A_c} \right)$$

where

- (A_d) = Differential Gain
- (A_c) = Common Mode Gain

Importance

A high CMRR ensures that only the difference between the two inputs affects the comparator output.

Common-mode noise originating from external interference is effectively rejected.

Practical Significance

High CMRR is essential in

- Biomedical instruments
- Industrial automation
- Aerospace electronics
- Precision measurement systems

Measurement Method

A common voltage is applied to both input terminals while maintaining a constant differential input.

The resulting shift in comparator switching threshold is measured and used to calculate CMRR.

3.6 Power Supply Rejection Ratio (PSRR)

Definition

Power Supply Rejection Ratio indicates the ability of the comparator to maintain consistent performance despite variations in supply voltage.

It is expressed as

$$PSRR = 20 \log \left(\frac{\Delta V_{Supply}}{\Delta V_{Offset}} \right)$$

Importance

Supply voltage fluctuations can influence the switching threshold.

A high PSRR indicates better immunity to power supply noise.

Measurement Method

The supply voltage is varied over a specified range while observing the corresponding change in input offset voltage.

The measured variation is used to determine PSRR.

3.7 Propagation Delay

Definition

Propagation delay is the time taken by the comparator output to respond after the input crosses the switching threshold.

It is one of the most important dynamic parameters of high-speed comparators.

Propagation delay determines the maximum operating speed of the comparator.

Types

- Low-to-High Delay (TPLH)
- High-to-Low Delay (TPHL)

Measurement

A square wave is applied to the comparator input.

Using the oscilloscope delay measurement function, the delay between the input transition and output transition is measured.

During the internship, the oscilloscope's automatic delay measurement feature was used to obtain these values.

Figure 4.3: Measurement of Propagation Delay (*Insert oscilloscope screenshot*)

3.8

3.9 Rise Time

Rise time is the time required for the output voltage to increase from **10% to 90%** of its final value.

Short rise time indicates fast switching capability.

3.10 Fall Time

Fall time is the time required for the output voltage to decrease from **90% to 10%** of its initial value.

Both rise time and fall time directly affect signal integrity in high-speed systems.

3.11 Differential Propagation Delay

In practical circuits, the propagation delay during a rising transition is not always equal to that during a falling transition.

The difference between these two delays is called the differential propagation delay.

Smaller differential delay improves timing accuracy.

3.12 Pulse Width

Pulse width is the duration for which the comparator output remains in a HIGH or LOW state.

Pulse width is important in timing analysis and digital interfacing.

3.13 Summary of Characterized Parameters

Parameter	Nature	Importance
V _{Trip+}	Static	Switching threshold (rising input)
V _{Trip-}	Static	Switching threshold (falling input)
Input Offset Voltage	Static	Comparator accuracy
CMRR	Static	Rejection of common-mode signals
PSRR	Static	Immunity to supply variations
Propagation Delay	Dynamic	Switching speed
Rise Time	Dynamic	Output transition speed
Fall Time	Dynamic	Output transition speed
Differential Delay	Dynamic	Timing symmetry
Pulse Width	Dynamic	Timing analysis

Table 3-1 Summary of Characterized Parameters

Chapter 4 : Experimental Setup and Laboratory Instrumentation

4.1 Introduction

The electrical characterization of high-speed analog integrated circuits requires a carefully designed experimental setup capable of generating precise input signals, providing stable power supplies, acquiring high-speed waveforms, and measuring electrical parameters with high accuracy. Since modern semiconductor devices operate at high frequencies and exhibit very small timing variations, manual measurements are often difficult and time-consuming. Therefore, automated laboratory instrumentation is widely employed to improve measurement accuracy, repeatability, and efficiency.

During this internship, the characterization of the SCL-designed high-speed comparator was carried out using an automated test setup consisting of a Digital Storage Oscilloscope (DSO), Arbitrary Function Generator (AFG), Source Measure Unit (SMU), Digital Multimeter (DMM), Arduino-controlled relay module, and a personal computer. The instruments were interconnected through USB interfaces and controlled using the Virtual Instrument Software Architecture (VISA) communication protocol. MATLAB served as the central platform for coordinating the measurement process and acquiring experimental data.

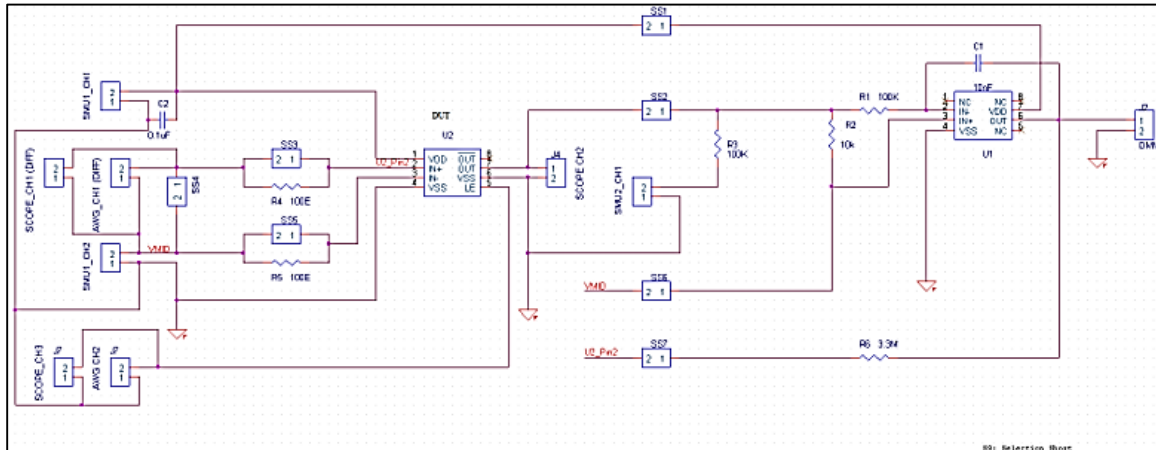
4.2 Experimental Setup

The complete characterization setup was designed to evaluate both the static and dynamic electrical parameters of the comparator. Each instrument performed a specific function within the measurement system.

The principal components of the setup included:

- SCL-designed High-Speed Comparator (Device Under Test)
- Digital Storage Oscilloscope
- Arbitrary Function Generator(AFG)
- Source Measure Unit(SMU)
- Digital Multimeter(DMM)
- Arduino-based Relay Switching Unit

- The computer acted as the central controller, communicating with each instrument through VISA interfaces. Appropriate commands were issued to configure the instruments, apply test conditions, and acquire measurement data automatically.



4.3 Device Under Test (DUT)

The comparator was mounted on a dedicated evaluation board that provided access to input terminals, output terminals, power supply pins, and latch enable signals. Proper decoupling and grounding practices were followed to ensure reliable measurements.

The Digital Storage Oscilloscope (DSO) was the primary instrument used for observing and measuring the input and output waveforms of the comparator. It enabled visualization of signal transitions and precise timing measurements required for propagation delay, rise time, fall time, and pulse width analysis.

The major functions performed by the oscilloscope during the internship included:

- Observation of input and output waveforms
- Measurement of propagation delay
- Measurement of rise time
- Measurement of fall time
- Pulse width measurement
- Verification of comparator switching behavior

Automatic measurement functions available in the oscilloscope were extensively used to improve measurement accuracy.



Figure 4.2 Digital Storage Oscilloscope Used for Characterization

4.5 Arbitrary Function Generator(model)

The Arbitrary Function Generator (AFG) was used to generate precise input signals required for comparator characterization.

During the experiments, two different waveforms were generated:

- The Arbitrary Function Generator generated **two synchronized signals** during characterization. A **100 Hz triangular waveform** was applied to the comparator input, while a **100 kHz square waveform** was applied to the **Latch Enable (LE)** input. The latch enable signal controlled the sampling operation of the comparator, allowing the output to update only at the active edges of the latch signal.
- A **square waveform** was applied during dynamic characterization to evaluate propagation delay, rise time, and fall time.

The latch enable terminal of the comparator was driven using a **100 kHz square wave**, enabling synchronized operation of the latched comparator.

The function generator allowed accurate adjustment of waveform amplitude, frequency, offset, and duty cycle according to the experimental requirements.

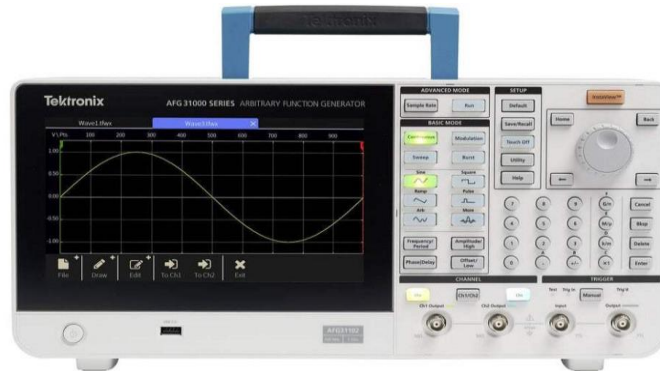


Figure 4.3Arbitrary Function Generator

4.6 Source Measure Unit (SMU)

The Source Measure Unit (SMU) combines the functions of a precision voltage/current source and a high-accuracy measurement instrument.

During characterization, the SMU was used to provide stable bias voltages and monitor the electrical behavior of the comparator under different supply conditions.

The SMU provides:

- Stable voltage sourcing
- Stable current sourcing
- Precision voltage measurement
- Precision current measurement

Its high accuracy makes it particularly suitable for semiconductor device characterization.



Figure 5.4: Source Measure Unit

4.7 Digital Multimeter (DMM)

The Digital Multimeter was used to verify voltage levels and ensure proper biasing of the comparator before characterization. It also served as an auxiliary instrument for validating measurements obtained from the automated setup.

Typical measurements included:

- DC voltage
- Resistance
- Continuity verification
- Supply voltage monitoring

The DMM contributed to ensuring the reliability and correctness of the overall measurement system.



Figure 4.4 Digital Multimeter

4.8 Arduino-Based Relay Switching Unit

To simplify the characterization process, an Arduino-controlled relay module was incorporated into the setup. The relay module enabled automatic switching between different measurement configurations without manual reconnection of the test setup.

The relay switching arrangement provided the following advantages:

- Reduced manual intervention
- Improved measurement repeatability
- Faster characterization
- Reduced possibility of wiring errors

The Arduino received commands from the host computer and controlled the relay positions accordingly.

4.9 VISA Communication

The laboratory instruments were connected to the personal computer through USB interfaces. Communication between the computer and the instruments was established using the Virtual Instrument Software Architecture (VISA), a standardized communication interface widely used in automated test and measurement systems.

VISA provides a common programming interface for controlling laboratory instruments from a computer, regardless of the instrument manufacturer.

The use of VISA allowed the characterization process to be automated by:

- Configuring instrument settings
- Initiating measurements
- Reading measured values
- Synchronizing multiple instruments
- Reducing manual errors

4.10 Automated Measurement Methodology

The characterization process followed a systematic sequence to ensure accurate and repeatable measurements.

The general measurement procedure consisted of the following steps:

1. Initialize all laboratory instruments.
2. Configure the oscilloscope, AFG, SMU according to the desired test conditions.
3. Apply the required input waveform to the comparator.
4. Apply the latch enable signal.
5. Acquire input and output waveforms.
6. Measure the required electrical parameter.
7. Record the measured values.
8. Repeat the process for all characterization parameters.

Automation significantly reduced the overall characterization time while ensuring consistency between successive measurements.

4.11 Experimental Precautions

To ensure reliable measurements, the following precautions were observed during characterization:

- Proper grounding was maintained throughout the setup.
- Instrument calibration was verified before measurements.
- Stable power supplies were used.
- Appropriate probe compensation was performed.
- Loose electrical connections were avoided.
- Measurement noise was minimized using proper shielding and cable management.
- Instrument settings were verified before each experiment.

4.12 Advantages of Automated Characterization

Compared to manual measurements, automated characterization offers several advantages:

- Higher measurement accuracy
- Reduced human error
- Improved repeatability
- Faster data acquisition
- Efficient handling of large datasets
- Simplified instrument control
- Consistent test conditions
- Reduced characterization time

These advantages make automation an essential component of modern semiconductor testing laboratories.

Chapter 5 : Characterization Results of the SCL designed High-Speed Comparator

5.1 Introduction

The characterization of integrated circuits is an essential stage in the semiconductor product development cycle. After fabrication, every integrated circuit must undergo comprehensive electrical testing to verify that its performance agrees with the intended design specifications. Electrical characterization provides valuable information regarding the static and dynamic behaviour of the device and helps identify the effects of fabrication process variations, supply voltage fluctuations, temperature changes, and other environmental factors.

High-speed comparators are widely used in analog-to-digital converters, communication systems, instrumentation, radar electronics, automotive control units, and aerospace applications. Since these applications demand high switching speed and excellent accuracy, the electrical performance of the comparator must be evaluated carefully before deployment. Parameters such as trip voltages, input offset voltage, propagation delay, common-mode rejection ratio, and power supply rejection ratio play a significant role in determining the overall performance of the comparator.

During this internship at the **VLSI Test and Application Development Division, Semiconductor Laboratory (SCL), Mohali**, the electrical characterization of an **SCL-designed high-speed comparator** was carried out using an automated laboratory setup. Unlike a conventional comparator that continuously monitors its inputs, the comparator used during this work is a **latched high-speed comparator**. The comparator performs the comparison operation only when the **Latch Enable (LE)** signal is active. Consequently, the output changes only at the specified edges of the latch enable signal, while remaining unchanged during the interval between two successive latch events.

The characterization process was performed using precision laboratory instruments, including a Digital Storage Oscilloscope (DSO), Arbitrary Function Generator (AFG), Source Measure Unit (SMU), Digital Multimeter (DMM), and an Arduino-controlled relay module. Communication between the computer and the instruments was established through the Virtual Instrument Software Architecture (VISA), allowing the characterization procedure to be automated. MATLAB was used to initialize the instruments, configure measurement conditions, acquire waveforms, and extract the required electrical parameters. Timing parameters such as propagation delay, rise time, and fall time were obtained using the automatic measurement functions available in the Tektronix oscilloscope.

The objective of this chapter is to describe the methodology adopted for characterizing the SCL-designed high-speed comparator and to explain the experimental procedures followed for measuring each electrical parameter.

5.2 Principle of Comparator Characterization

The characterization methodology adopted during this internship was based on synchronized sampling using a latch enable signal. Since the comparator is a latched device, the comparison between the input voltage and the reference voltage is performed only at the active edges of the latch enable signal.

A low-frequency triangular waveform was applied to the comparator input. The triangular waveform changes gradually with time, allowing the switching threshold of the comparator to be determined accurately. Simultaneously, a high-frequency square wave was applied to the latch enable input. At every active edge of the latch signal, the comparator samples the instantaneous value of the input voltage and updates its output accordingly. Between two consecutive latch events, the output remains latched and does not change even though the input voltage continues to vary.

This method provides an accurate means of determining the switching thresholds of the comparator while ensuring synchronized operation during characterization.

The operating conditions used during the characterization are summarized in Table 6.1.

Table 5-1 Operating Conditions Used During Characterization

Parameter	Value
Device Under Test	SCL-designed High-Speed Comparator
Input Waveform	Triangular Wave
Input Frequency	100 Hz
Latch Enable Waveform	Square Wave
Latch Enable Frequency	100 kHz
Measurement Mode	Automated Characterization

5.3 Experimental Setup

The characterization setup was designed to facilitate accurate measurement of both static and dynamic electrical parameters of the comparator. The setup comprised precision laboratory instruments interconnected through a personal computer. The computer acted as the central controller, issuing commands to the instruments and collecting the measured data.

The major components of the characterization setup are listed below:

- SCL-designed High-Speed Comparator (Device Under Test)
- Digital Storage Oscilloscope (Tektronix)
- Arbitrary Function Generator (Tektronix)
- Source Measure Unit
- Arduino-controlled Relay Module
- Personal Computer

The comparator was mounted on a dedicated evaluation board that provided access to all necessary input, output, power supply, and latch enable connections. Proper grounding and shielding were maintained throughout the experimental setup to minimize electrical noise and ensure reliable measurements.

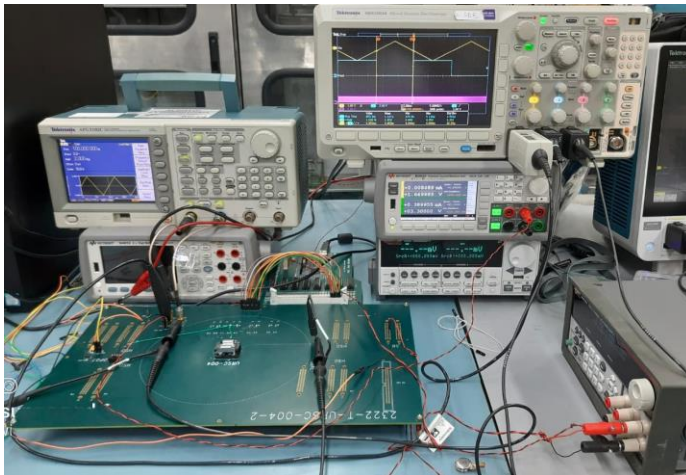


Figure 5.1 Experimental characterization setup

5.4 Signal Generation

The Arbitrary Function Generator was configured to generate the signals required for characterization. Two independent waveforms were used during the experiments.

Comparator Input Signal

A **100 Hz triangular waveform** was applied to the comparator input terminal. The slowly varying input ensured that the switching threshold of the comparator could be identified accurately. As the triangular waveform increased and decreased linearly, the comparator sampled different input voltage levels at successive latch events.

Latch Enable Signal

A **100 kHz square waveform** was applied to the latch enable input of the comparator. The latch enable signal controlled the sampling operation of the comparator. At every active edge of the latch enable signal, the comparator evaluated the instantaneous input voltage and updated its output. Between two consecutive latch edges, the output remained unchanged regardless of the variation in the input waveform.

The combination of a slowly varying triangular input and a high-frequency latch signal enabled precise measurement of the comparator switching thresholds while maintaining synchronized operation.

5.5 Instrument Configuration

Before beginning the characterization process, all laboratory instruments were configured according to the required operating conditions.

The Arbitrary Function Generator was programmed to generate the specified input and latch waveforms. The Source Measure Unit supplied the required operating voltage to the comparator and monitored the supply conditions throughout the experiment. The Digital Multimeter was used to verify the applied voltages before starting the measurements.

The Digital Storage Oscilloscope was configured to simultaneously display the input waveform, latch enable signal, and comparator output. Appropriate voltage scales, time bases, trigger settings, and probe compensation were selected to obtain stable and noise-free waveforms.

Once the instruments were configured, communication between the computer and each instrument was verified through the VISA interface to ensure successful automation of the characterization process.

5.6 Automation of the Characterization Process

To improve efficiency and eliminate manual intervention, the characterization process was automated using MATLAB. Rather than manually adjusting each instrument for every measurement, the software coordinated the operation of the complete measurement setup.

The automation procedure involved the following operations:

- Initializing all laboratory instruments.
- Establishing communication through the VISA interface.
- Configuring the Arbitrary Function Generator.
- Configuring the Digital Storage Oscilloscope.
- Setting the required operating conditions.
- Controlling the Arduino relay module.
- Acquiring measurement data.
- Extracting the required electrical parameters.
- Saving the measurement results for further analysis.

Automation significantly reduced characterization time and improved repeatability by ensuring that identical measurement conditions were maintained throughout the experiments.

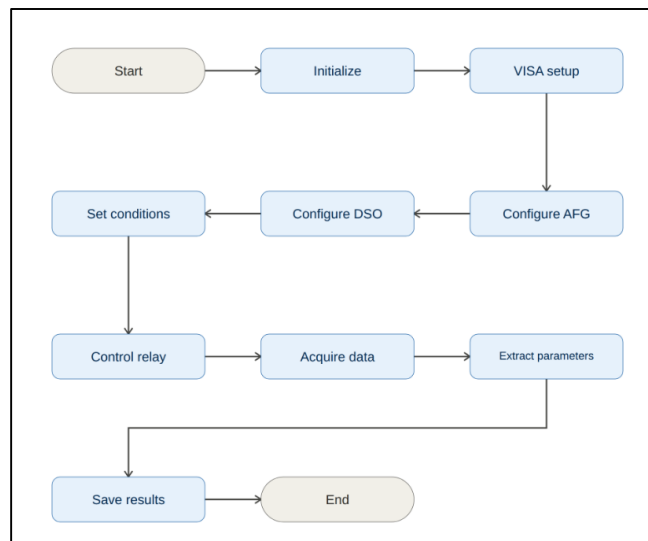


Figure 5.2 Flowchart of the automated characterization process

5.7 General Measurement Procedure

The electrical characterization of the comparator followed a systematic sequence to ensure accurate and repeatable measurements.

The procedure adopted during the internship is summarized below:

1. The Device Under Test (DUT) was connected to the characterization setup.
2. All laboratory instruments were powered on and initialized.
3. Communication between the computer and the instruments was established through VISA.
4. The Source Measure Unit supplied the required operating voltage to the comparator.
5. A 100 Hz triangular waveform was applied to the comparator input.
6. A 100 kHz square waveform was applied to the latch enable input.
7. The oscilloscope simultaneously displayed the input waveform, latch enable waveform, and comparator output waveform.
8. The comparator sampled the input voltage at every active edge of the latch enable signal.
9. The corresponding electrical parameter was measured using either automated data acquisition or the oscilloscope's built-in measurement functions.
10. The measured values were recorded for further analysis.

This sequence was repeated for each electrical parameter evaluated during the characterization process.

5.8 Measurement of Positive Trip Voltage (V_{Trip+})

5.8.1 Objective

The objective of this experiment was to determine the **Positive Trip Voltage (V_{Trip+})** of the SCL-designed high-speed comparator. Positive Trip Voltage is defined as the input voltage at which the comparator output changes from one logic state to the other while the input voltage is increasing.

This parameter is one of the most important static characteristics of a comparator because it represents the upper switching threshold of the device.

5.8.2 Principle

In an ideal comparator, the output changes state exactly when the differential input voltage becomes zero. However, due to unavoidable mismatches in transistor dimensions, threshold voltages, fabrication tolerances, and process variations, the practical switching point deviates slightly from the ideal value.

Since the comparator characterized during this internship is a **latched comparator**, the output does not change continuously with the input. Instead, the comparator samples the input voltage only at the active edges of the latch enable signal.

As the triangular waveform gradually increases, the comparator continues sampling the instantaneous input voltage at every latch event. Once the sampled input exceeds the internal switching threshold, the comparator updates its output. The input voltage corresponding to this transition is defined as the **Positive Trip Voltage (V_{Trip+})**.

5.8.3 Experimental Procedure

The characterization setup described in the previous section was used for measuring the positive trip voltage.

Initially, the comparator was supplied with the required operating voltage using the Source Measure Unit. A **100 Hz triangular waveform** generated by the Arbitrary Function Generator was applied to the comparator input terminal. Simultaneously, a **100 kHz square waveform** was applied to the Latch Enable (LE) input.

The Digital Storage Oscilloscope displayed three signals simultaneously:

- Input triangular waveform
- Latch Enable waveform
- Comparator output waveform

During the rising portion of the triangular waveform, the comparator sampled the input voltage at every active edge of the latch enable signal. As soon as the sampled input exceeded the comparator threshold, the output changed to the opposite logic state.

The corresponding input voltage at that sampling instant was identified as the **Positive Trip Voltage**.

The measured waveform was acquired automatically through the characterization software, while the switching point was extracted from the captured data.

5.8.4 Observation

It was observed that the comparator output remained unchanged between two successive latch events. The output transition occurred only at the active edge of the latch enable signal after the sampled input voltage exceeded the internal switching threshold.

This behaviour confirms the proper operation of the latch circuitry incorporated within the comparator.

5.8.5 Importance

Positive Trip Voltage determines the switching threshold of the comparator during an increasing input signal.

Accurate measurement of this parameter is important because it directly affects:

- Comparator accuracy
- Analog-to-digital conversion accuracy
- Threshold detection
- Signal integrity
- Overall system performance

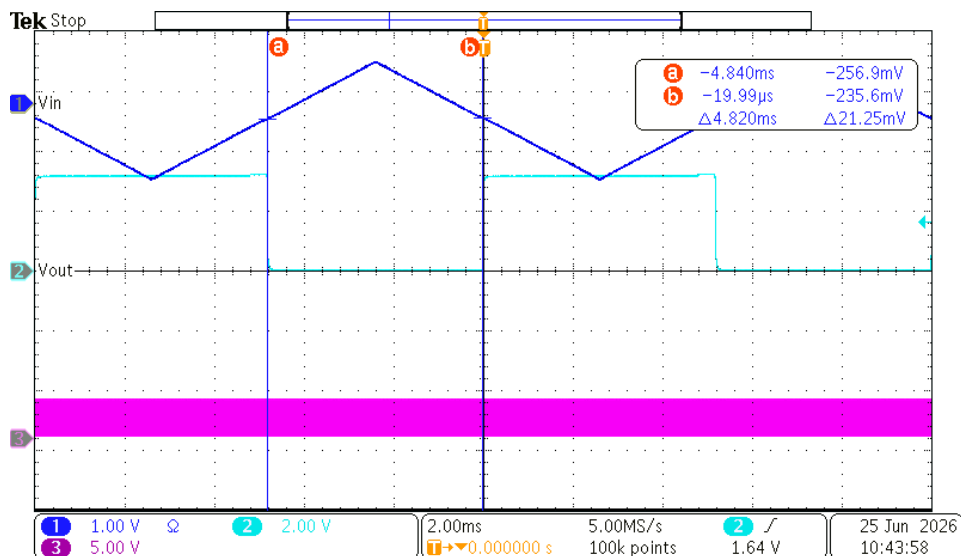


Figure 5.3 Measurement of Positive Trip Voltage

5.9 Measurement of Negative Trip Voltage (V_{Trip-})

5.9.1 Objective

The objective of this experiment was to determine the **Negative Trip Voltage (V_{Trip-})** of the comparator.

Negative Trip Voltage is defined as the input voltage at which the comparator output changes state while the input voltage is decreasing.

5.9.2 Principle

As the triangular waveform decreases, the comparator continues sampling the instantaneous input voltage at every active edge of the latch enable signal.

When the sampled input voltage falls below the comparator switching threshold, the output changes to the opposite logic level.

The corresponding input voltage is called the **Negative Trip Voltage (V_{Trip-})**.

5.9.3 Experimental Procedure

The same experimental setup used for V_{Trip+} measurement was employed.

The triangular waveform was allowed to decrease continuously while the comparator sampled the input voltage at successive latch events.

The oscilloscope simultaneously displayed:

- Input waveform
- Latch Enable waveform
- Comparator output waveform

During the falling portion of the triangular waveform, the comparator output changed its logic state when the sampled input voltage became lower than the switching threshold.

The corresponding input voltage was recorded as the Negative Trip Voltage.

5.9.4 Observation

The comparator output remained constant between two latch enable events and changed state only after the active edge of the latch enable signal.

This confirmed that the comparator was operating in a latched mode rather than as a continuously operating comparator.

5.9.5 Importance

Negative Trip Voltage is an important parameter because it determines the switching threshold during decreasing input conditions.

Together with the positive trip voltage, it provides information regarding:

- Comparator offset
- Comparator hysteresis
- Switching accuracy
- Device matching

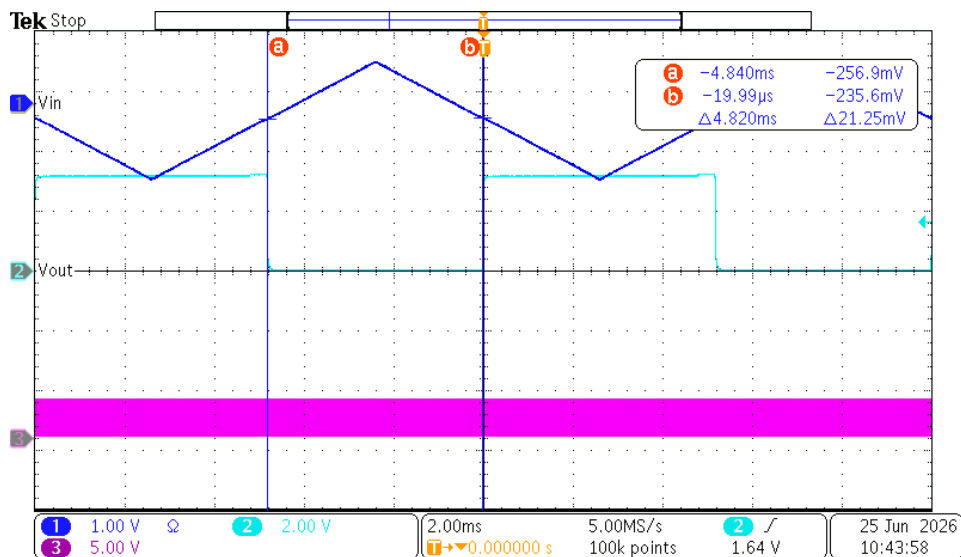


Figure 5.4 Measurement of Negative Trip Voltage

5.10 Measurement of Input Offset Voltage (VIO)

5.10.1 Objective

The objective of this experiment was to determine the inherent input offset voltage of the comparator.

5.10.2 Principle

Ideally, a comparator should switch exactly when both input terminals are at the same potential.

However, practical comparators exhibit a small input-referred error known as **Input Offset Voltage (VIO)** due to transistor mismatch and process variations.

After obtaining the positive and negative trip voltages, the input offset voltage was determined using

$$VIO = \frac{V_{Trip+} + V_{Trip-}}{2}$$

where

- V_{Trip+} = Positive Trip Voltage
- V_{Trip-} = Negative Trip Voltage

5.10.3 Experimental Procedure

The positive and negative trip voltages obtained from the previous measurements were used to determine the input offset voltage.

The acquired waveform data were processed automatically, and the corresponding offset voltage was calculated using the above equation.

No additional experimental setup was required because the offset voltage was derived directly from the measured switching thresholds.

5.10.4 Observation

The measured input offset voltage represented the deviation of the practical switching threshold from the ideal switching point.

The offset originates primarily from:

- Process variation
- Device mismatch
- Threshold voltage variation
- Layout asymmetry
- Temperature effects

5.10.5 Importance

Input offset voltage is one of the most important performance parameters of a comparator.

A smaller input offset voltage indicates:

- Better matching of internal devices
- Higher comparator accuracy
- Improved precision
- Better overall performance

5.11 Measurement of Comparator Hysteresis

5.11.1 Objective

To determine the hysteresis of the comparator.

5.11.2 Principle

Comparator hysteresis is defined as the difference between the positive and negative trip voltages. It is given by

$$V_H = V_{Trip+} - V_{Trip-}$$

Hysteresis prevents false switching caused by electrical noise around the threshold voltage.

5.11.3 Experimental Procedure

After measuring both trip voltages, the hysteresis was determined by calculating the difference between the measured values.

Since the required data were already available from previous measurements, no additional experiment was necessary.

5.11.4 Observation

The measured hysteresis represented the difference between the switching thresholds corresponding to increasing and decreasing input voltages.

5.11.5 Importance

Comparator hysteresis improves the overall robustness of the comparator by:

- Preventing output oscillations
- Rejecting noise
- Improving switching stability
- Reducing false triggering
- Increasing system reliability

5.12 Measurement of Common Mode Rejection Ratio (CMRR)

5.12.1 Objective

The objective of this experiment was to evaluate the **Common Mode Rejection Ratio (CMRR)** of the SCL-designed high-speed comparator. CMRR indicates the ability of the comparator to reject signals that are simultaneously present at both input terminals.

A high value of CMRR ensures that only the differential component of the input signal influences the comparator output while common-mode disturbances are effectively rejected.

5.12.2 Principle

In practical electronic systems, both input terminals of a comparator may experience unwanted noise due to electromagnetic interference, ground potential differences, or coupling from nearby circuits. Since these unwanted voltages appear simultaneously on both inputs, they are referred to as **common-mode signals**.

An ideal comparator completely rejects these common-mode signals and responds only to the difference between the two input voltages. Practical comparators exhibit a finite common-mode gain; therefore, their ability to reject common-mode signals is expressed using the Common Mode Rejection Ratio.

Mathematically,

$$CMRR = 20 \log_{10} \left(\frac{A_d}{A_c} \right)$$

where

- A_d = Differential Gain
- A_c = Common Mode Gain

A higher value of CMRR indicates better immunity to common-mode noise.

5.12.3 Experimental Procedure

During characterization, the comparator was operated under the required bias conditions while maintaining the differential input. The common-mode voltage was varied within the specified operating range, and the corresponding shift in the switching threshold was observed.

The characterization setup consisted of:

- Source Measure Unit
- Arbitrary Function Generator
- Digital Storage Oscilloscope

- Comparator Evaluation Board

The measured values were processed through the automated characterization setup to determine the variation in switching threshold resulting from the applied common-mode voltage.

5.12.4 Observation

It was observed that the comparator exhibited only a small variation in switching threshold with changes in common-mode voltage, indicating satisfactory common-mode rejection capability.

5.12.5 Importance

A high Common Mode Rejection Ratio is essential for applications such as

- High-speed data acquisition systems
- Precision instrumentation
- Communication receivers
- Industrial measurement systems
- Aerospace electronics

where electrical noise is unavoidable.

5.13 Measurement of Power Supply Rejection Ratio (PSRR)

5.13.1 Objective

The objective of this experiment was to evaluate the influence of supply voltage variations on comparator performance.

5.13.2 Principle

Power supply fluctuations may alter the operating conditions of analog integrated circuits and affect their switching characteristics. The ability of a comparator to maintain a stable switching threshold despite variations in supply voltage is expressed using the **Power Supply Rejection Ratio (PSRR)**.

A comparator with high PSRR exhibits minimal change in its operating characteristics when the supply voltage varies within its specified range.

5.13.3 Experimental Procedure

The Source Measure Unit supplied the operating voltage to the comparator.

The supply voltage was varied over the required operating range while monitoring the switching threshold of the comparator.

The corresponding change in comparator characteristics was recorded using the automated characterization system.

5.13.4 Observation

The measured switching threshold exhibited only a small variation as the supply voltage changed, indicating satisfactory immunity against supply voltage fluctuations.

5.13.5 Importance

High PSRR is important because it

- Improves system reliability
- Reduces the influence of supply noise
- Maintains comparator accuracy
- Enhances performance in noisy environments

5.14 Measurement of Propagation Delay

5.14.1 Objective

The objective of this experiment was to determine the propagation delay of the SCL-designed high-speed comparator.

Propagation delay is one of the most important dynamic parameters because it determines the response speed of the comparator after an input event.

5.14.2 Principle

Propagation delay is defined as the time interval between the occurrence of a triggering event and the corresponding transition at the comparator output.

Since the comparator used during this internship is a **latched comparator**, the output transition is synchronized with the **Latch Enable (LE)** signal. The comparator samples the input at the active latch edge and updates its output after a finite internal delay. Therefore, the measured propagation delay represents the time required by the comparator to process the sampled input and produce the corresponding output transition.

5.14.3 Experimental Procedure

Unlike the measurements of static parameters, propagation delay was evaluated by applying a **square-wave input signal** to the comparator.

The latch enable signal remained a **100 kHz square wave**, ensuring synchronized operation of the comparator.

The Digital Storage Oscilloscope was configured to measure the delay automatically using its built-in timing measurement function.

The measurement settings used during the experiment were:

- **Source A: Channel 3**
- **Source B: Channel 2**

Different edge combinations were selected depending on the parameter being measured. The oscilloscope automatically calculated the delay between the selected transition of the reference signal and the corresponding transition of the comparator output.

The measured delay values were displayed directly by the oscilloscope and recorded for analysis.

5.14.4 Measurement of Low-to-High Propagation Delay (TPLH)

The **Low-to-High Propagation Delay (TPLH)** represents the time required for the comparator output to transition from LOW to HIGH after the selected reference edge.

The oscilloscope delay measurement function was configured to measure the delay between the appropriate reference edge and the corresponding rising transition of the output waveform.

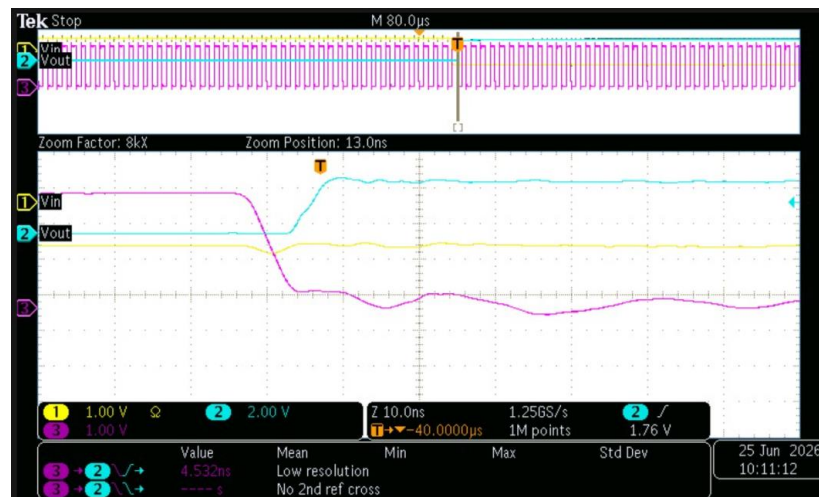


Figure 5.5 Measurement of TPLH

5.14.5 Measurement of High-to-Low Propagation Delay (TPHL)

The **High-to-Low Propagation Delay (TPHL)** represents the time required for the comparator output to transition from HIGH to LOW after the selected reference edge.

The oscilloscope automatically measured the delay between the selected reference transition and the corresponding falling edge of the comparator output.

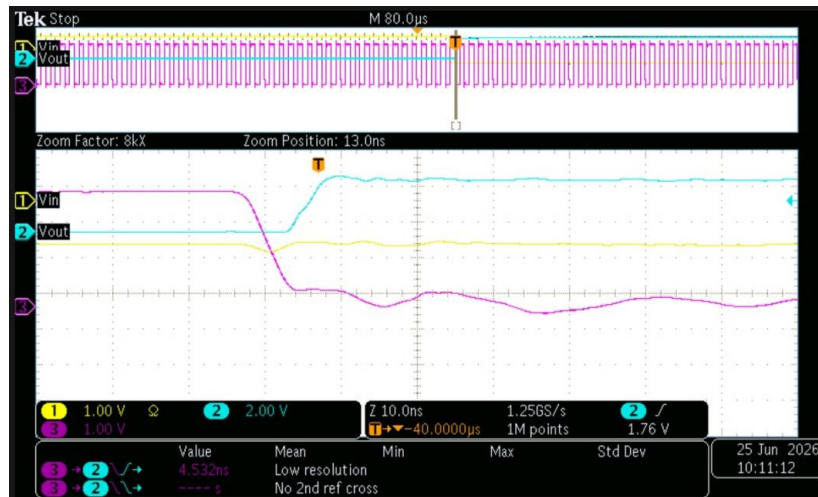


Figure 5.6 Measurement of TPHL

5.14.6 Observation

The oscilloscope displayed the propagation delay directly using its automatic timing measurement feature.

Separate delay values were obtained for different edge combinations by selecting the required source channels and transition types.

The use of the automatic measurement function significantly improved measurement accuracy and eliminated manual timing errors.

5.15 Measurement of Rise Time

5.15.1 Objective

The objective of this experiment was to determine the **rise time** of the comparator output waveform. Rise time is an important dynamic parameter that indicates how quickly the comparator output changes from the LOW logic level to the HIGH logic level after switching.

A smaller rise time indicates a faster output transition and better suitability for high-speed applications.

5.15.2 Principle

Rise time is defined as the time required for the output voltage to increase from **10% to 90%** of its final steady-state value.

For high-speed comparators, rise time depends on several factors including:

- Internal output stage design
- Load capacitance
- Supply voltage
- Device technology
- Output drive capability

Since the comparator output changes only after the latch samples the input signal, the rise time represents the transition speed of the output after the switching decision has already been made.

5.15.3 Experimental Procedure

For rise time measurement, a square-wave signal was applied to the comparator input while the latch enables continued operating at the specified frequency.

The comparator output waveform was displayed on the Digital Storage Oscilloscope.

The oscilloscope's automatic measurement function was used to determine the rise time. The instrument automatically measured the interval between the **10%** and **90%** voltage levels of the output waveform.

The measured value was displayed directly on the oscilloscope and recorded for further analysis.

5.15.4 Observation

The oscilloscope accurately determined the rise time of the comparator output using its built-in measurement feature.

The measured value represents the speed at which the output driver charges the output node during the LOW-to-HIGH transition.

5.15.5 Importance

Rise time affects the maximum operating speed of digital systems.

A smaller rise time offers the following advantages:

- Faster switching
- Improved signal integrity
- Higher operating frequency
- Reduced timing uncertainty

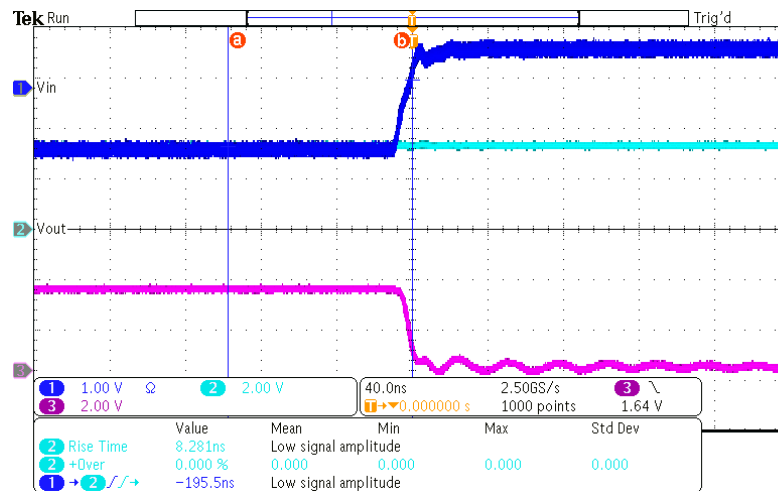


Figure 5.7 Measurement of Rise Time

5.16 Measurement of Fall Time

5.16.1 Objective

The objective of this experiment was to determine the **fall time** of the comparator output waveform.

5.16.2 Principle

Fall time is defined as the time required for the output voltage to decrease from **90%** to **10%** of its initial value.

Together, rise time and fall time describe the dynamic performance of the comparator output stage.

5.16.3 Experimental Procedure

The comparator output waveform was observed using the Digital Storage Oscilloscope.

The oscilloscope's automatic measurement function was configured to measure the fall time directly from the output waveform.

The instrument automatically calculated the interval between the **90%** and **10%** voltage levels.

5.16.4 Observation

The measured fall time represents the speed at which the comparator output discharges during the HIGH-to-LOW transition.

5.16.5 Importance

Low fall time improves:

- High-speed operation
- Digital timing performance
- Communication speed
- Overall comparator performance

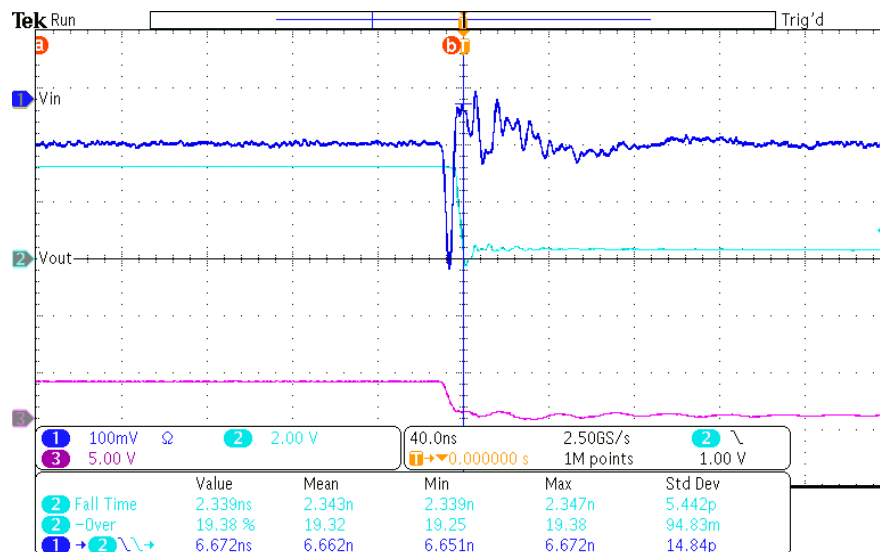


Figure 5.8 Measurement of Fall Time

5.17 Automated Measurement Procedure

To improve measurement efficiency and eliminate manual intervention, the characterization process was automated using MATLAB.

The software established communication with all laboratory instruments through the VISA interface and coordinated their operation throughout the characterization process.

The automated procedure consisted of the following steps:

1. Initialize all laboratory instruments.
2. Establish VISA communication.
3. Configure the Arbitrary Function Generator.
4. Configure the Source Measure Unit.
5. Configure the Digital Storage Oscilloscope.
6. Apply the required input waveform.
7. Apply the latch enable waveform.
8. Acquire the input, latch enable, and output waveforms.
9. Extract the required electrical parameters.
10. Save the measured data.

DeviceID	Steady_cu	InputOffset	V_plus	V_minus	CMRR	PSRR	FallTime	NegShoot	HL_Delay	RiseTime	PosShoot	LH_Delay	Diff_Delay	PulseWidth	Date_Time
1	0.000393	-215.522	-215.973	-215.07	45.83264	36.98104	4.80E-09	0.00E+00	-2.75E-09	8.09E-09	0.00E+00	-9.21E-09	6.46E-09	6E-09	#####
2	0.000406	-220.013	-221.007	-219.018	47.98756	36.52808	7.14E-09	0.00E+00	-5.74E-10	7.97E-10	0.00E+00	-3.43E-09	2.85E-09	6E-09	#####
3	0.000392	-222.63	-224.241	-221.02	45.68408	37.50066	5.86E-09	0.00E+00	-3.70E-09	3.15E-08	0.00E+00	-1.80E-08	1.43E-08	6E-09	#####
4	0.0004	-208.078	-208.953	-207.203	47.62932	35.14053	5.93E-09	0.00E+00	5.69E-09	4.67E-09	0.00E+00	-3.98E-09	9.68E-09	6E-09	#####
5	0.00037	-216.276	-220.043	-212.509	79.78009	34.36758	8.70E-09	0.00E+00	-2.56E-09	6.29E-09	0.00E+00	-2.14E-09	4.29E-10	6E-09	#####

Figure 5.9 Measured data after automatically saving into Excel file

Automation significantly reduced characterization time while improving measurement repeatability and minimizing operator-dependent errors.

5.18 Experimental Precautions

Several precautions were observed during characterization to ensure accurate and repeatable measurements.

- All laboratory instruments were allowed to stabilize before beginning the experiment.

- The comparator was operated within its specified supply voltage range.
- Proper grounding was maintained throughout the experimental setup.
- Probe compensation of the oscilloscope was verified before measurements.
- Loose electrical connections were avoided.
- The signal cables were kept as short as possible to minimize parasitic effects.
- Instrument calibration was verified before testing.
- Stable supply voltages were maintained using the Source Measure Unit.
- Appropriate trigger settings were selected on the oscilloscope to obtain stable waveforms.
- The measurement setup was checked before recording experimental observations.

Following these precautions improved the reliability and repeatability of the characterization results.

5.19 Sources of Measurement Uncertainty

Although the characterization setup was carefully designed, minor uncertainties may arise due to various practical factors.

The principal sources of measurement uncertainty include:

- Instrument resolution
- Oscilloscope timing accuracy
- Probe loading
- Cable parasitic
- Electrical noise
- Supply voltage fluctuations
- Temperature variations
- Semiconductor process variations

These effects were minimized through proper instrument calibration, shielding, grounding, and automation of the measurement procedure.

Chapter 6 : Study of Radiation Testing of High Speed Comparator

6.1 Introduction

Radiation qualification is an essential aspect of semiconductor device development for applications operating in harsh radiation environments such as space missions, defence electronics, nuclear instrumentation, and high-energy physics experiments. Electronic components deployed in these environments are continuously exposed to ionizing radiation, which can alter their electrical characteristics and degrade circuit performance over time. Consequently, semiconductor devices intended for such applications must undergo radiation testing to evaluate their reliability and ensure compliance with established qualification standards.

High-speed comparators are critical building blocks in Analog-to-Digital Converters (ADCs), communication systems, instrumentation, radar electronics, and aerospace applications. Since these devices operate at high switching speeds and require precise threshold detection, even small radiation-induced changes in parameters such as input offset voltage, propagation delay, or supply current may significantly affect overall system performance. Radiation qualification therefore plays an important role in validating the suitability of high-speed comparators for radiation environments.

During this internship at the **VLSI Test and Application Development Division, Semiconductor Laboratory (SCL), Mohali**, a study was undertaken to understand the methodology used for radiation qualification of high-speed comparators. This study focused on **Biased High Dose Rate (HDR) Total Ionizing Dose (TID) testing**, which is one of the standard methods used to evaluate radiation tolerance in semiconductor devices.

The study was primarily based on the following technical references:

- **Texas Instruments, "TLVIH103 High-Speed Comparator Radiation Test Report,"** which describes the radiation qualification of the TLVIH103 comparator using Biased High Dose Rate testing.
- **MIL-STD-883 Test Method 1019.9, Condition A**, which specifies the standard procedure for Total Ionizing Dose testing of microelectronic devices.
- **Texas Instruments, "TLV360x 325-MHz High-Speed Comparator Datasheet,"** which was referred to for understanding comparator electrical parameters and their significance before and after radiation exposure.

Although radiation exposure experiments were **not performed** during the internship, studying these documents provided valuable understanding of industrial radiation qualification procedures, radiation test setups, biasing requirements during irradiation, and the electrical characterization carried out before and after radiation exposure.

This chapter presents the concepts learned during the study and explains the methodology adopted for Biased High Dose Rate testing in accordance with the procedures described in the referenced technical documents.

6.2 Radiation Environment and the Need for Radiation Testing

Electronic systems designed for terrestrial applications generally operate under normal environmental conditions and are seldom exposed to significant levels of ionizing radiation. However, electronic devices used in satellites, launch vehicles, deep-space probes, defence equipment, avionics, and nuclear facilities operate in environments where radiation exposure is unavoidable.

The principal sources of radiation affecting semiconductor devices include:

- Galactic Cosmic Rays (GCR)
- Solar Particle Events (SPE)
- Trapped particles in the Van Allen radiation belts
- Gamma radiation
- X-rays
- High-energy protons
- Heavy ions
- Neutrons

These energetic particles interact with semiconductor materials and may introduce permanent or temporary changes in device characteristics. Such changes can reduce circuit reliability and eventually lead to functional failure if appropriate radiation-hardening techniques are not employed.

For high-speed comparators, radiation-induced degradation can influence critical electrical parameters including:

- Input offset voltage

- Propagation delay
- Supply current
- Switching threshold
- Leakage current
- Output logic levels

Consequently, radiation testing is carried out to verify that these parameters remain within specified limits after exposure to ionizing radiation.

6.3 Total Ionizing Dose (TID) Effects

Among the various radiation effects encountered in semiconductor devices, **Total Ionizing Dose (TID)** is one of the most important long-term degradation mechanisms.

According to **MIL-STD-883 Test Method 1019.9**, Total Ionizing Dose represents the cumulative amount of ionizing radiation absorbed by a semiconductor device during its operational lifetime. Exposure to ionizing radiation generates electron-hole pairs within the oxide layers of MOS devices. While many of these charge carriers recombine, a fraction becomes trapped in the gate oxide and at the silicon-oxide interface. The accumulation of trapped charge gradually modifies the electrical characteristics of the transistors forming the integrated circuit.

As discussed in the **TLVIH103 High-Speed Comparator Radiation Test Report**, radiation-induced oxide charge accumulation may influence several comparator parameters, including the input offset voltage, propagation delay, leakage current, and supply current. These changes become more pronounced as the absorbed radiation dose increases.

For this reason, TID testing is performed by measuring the electrical characteristics of the device before irradiation and comparing them with measurements obtained after exposure to specified radiation doses.

The overall objective of TID testing is to determine whether the device continues to satisfy its electrical specifications throughout the required radiation dose range.

6.4 Types of Radiation Effects on Semiconductor Devices

Radiation-induced degradation in semiconductor devices is generally classified into three categories:

6.4.1 Total Ionizing Dose (TID)

Total Ionizing Dose effects result from the cumulative absorption of ionizing radiation over time. These effects primarily influence oxide layers and semiconductor interfaces, producing gradual degradation of electrical characteristics.

Typical manifestations include:

- Threshold voltage shift
- Increased leakage current
- Increased propagation delay
- Input offset voltage variation
- Increased power consumption

6.4.2 Single Event Effects (SEE)

Single Event Effects occur when a single energetic particle strikes a sensitive region of the integrated circuit. Unlike TID effects, SEE is an instantaneous phenomenon and may produce transient or permanent failures.

Common examples include:

- Single Event Upset (SEU)
- Single Event Transient (SET)
- Single Event Latch-up (SEL)
- Single Event Burnout (SEB)
- Single Event Gate Rupture (SEGR)

6.4.3 Displacement Damage (DD)

Displacement damage occurs when energetic particles physically displace atoms from the semiconductor crystal lattice. The resulting defects reduce carrier lifetime and mobility, degrading overall device performance.

Typical effects include:

- Increased series resistance
- Reduced transistor gain
- Increased leakage current
- Reduced switching speed
- Lower device reliability

6.5 Biased High Dose Rate (HDR) Testing

One of the most widely adopted techniques for evaluating the radiation tolerance of semiconductor devices is **Biased High Dose Rate (HDR) Total Ionizing Dose (TID) testing**. This method evaluates the electrical performance of a device while it is exposed to ionizing radiation under specified operating bias conditions. Maintaining electrical bias during irradiation allows the test to closely simulate the actual operating conditions encountered by semiconductor devices in practical applications.

During this internship, the methodology of **Biased High Dose Rate testing** was studied with reference to the **TLVIH103 High-Speed Comparator Radiation Test Report** [2]. The report illustrates the qualification procedure adopted for evaluating the radiation tolerance of a high-speed comparator using standardized irradiation techniques.

Unlike unbiased irradiation, where the device remains unpowered during exposure, biased irradiation allows electric fields to exist across the internal oxide layers of MOS transistors. These electric fields influence the transport and trapping of radiation-generated charge carriers, thereby producing degradation mechanisms that closely resemble those occurring in real operating environments [3].

The primary objective of biased HDR testing is to determine whether the electrical characteristics of the comparator remain within the specified limits after exposure to the required radiation dose.

Typical parameters monitored before and after irradiation include:

- Input Offset Voltage
- Positive and Negative Trip Voltages
- Supply Current
- Propagation Delay
- Rise Time
- Fall Time
- Output Logic Levels
- Functional Operation

The measured values are compared with their pre-irradiation values to evaluate radiation-induced degradation.

6.6 MIL-STD-883 Test Method 1019.9 – Condition A

To ensure consistency and repeatability in radiation qualification, semiconductor manufacturers follow internationally recognized testing standards. One of the most widely accepted standards for Total Ionizing Dose testing is **MIL-STD-883 Test Method 1019.9** [3].

This standard establishes a uniform procedure for evaluating the effects of ionizing radiation on microelectronic devices intended for military, aerospace, and other high-reliability applications.

The study conducted during the internship focused specifically on **Condition A** of Method 1019.9, which specifies the requirements for **steady-state irradiation at a controlled dose rate** while maintaining the prescribed electrical bias conditions.

The standard defines several important aspects of radiation qualification, including:

- Radiation source
- Dose rate
- Total accumulated dose
- Electrical bias conditions
- Dosimetry requirements

- Measurement intervals
- Annealing procedure (where applicable)
- Post-irradiation electrical characterization
- Pass/fail acceptance criteria

By following a standardized testing procedure, the results obtained from different laboratories become directly comparable, thereby improving the reliability of radiation qualification.

The **TLVIH103 Radiation Test Report** follows this methodology while evaluating the radiation tolerance of the comparator [2].

6.7 Radiation Test Facility

Radiation testing is generally carried out in specialized irradiation facilities equipped with calibrated gamma radiation sources and precision dosimetry systems.

For Total Ionizing Dose testing, **Cobalt-60 (Co-60)** is one of the most commonly used gamma radiation sources because of its stable energy spectrum and well-characterized radiation output [3].

A typical radiation test facility consists of:

- Cobalt-60 gamma source
- Irradiation chamber
- Device mounting fixture
- Bias supply arrangement
- Dosimetry system
- Radiation shielding
- Control room
- Electrical measurement equipment

The semiconductor device under test is mounted inside the irradiation chamber while maintaining the specified electrical bias conditions. Radiation exposure is then performed until the required absorbed dose is achieved.

The absorbed radiation dose is monitored using calibrated dosimeters to ensure compliance with the prescribed irradiation conditions.

6.8 Bias Conditions During Irradiation

An important aspect of Biased High Dose Rate testing is maintaining appropriate electrical bias throughout the irradiation process.

The **TLVIH103 High-Speed Comparator Radiation Test Report** specifies the required bias conditions during irradiation so that the device experiences electrical stress similar to actual operating conditions [2].

Applying electrical bias during irradiation affects the movement and trapping of radiation-generated charge carriers inside the oxide layers. Consequently, the degradation observed under biased conditions is generally considered more representative of practical operation than that obtained from unbiased testing.

Typical bias conditions include:

- Specified supply voltage
- Input bias conditions
- Output loading
- Ground reference
- Device operating mode

These bias conditions are maintained continuously throughout irradiation unless otherwise specified by the applicable qualification procedure.

Maintaining stable bias conditions ensures repeatability of the experiment and allows meaningful comparison between pre-irradiation and post-irradiation electrical characteristics.

6.9 Dosimetry

An essential requirement of radiation testing is the accurate measurement of the absorbed radiation dose.

The process of determining the absorbed dose is known as **dosimetry**.

According to **MIL-STD-883 Test Method 1019.9** [3], calibrated dosimeters are placed at appropriate locations within the irradiation chamber to verify the radiation dose received by the Device Under Test (DUT).

The dosimetry system provides information regarding:

- Dose rate
- Total accumulated dose
- Radiation uniformity
- Irradiation time

Accurate dosimetry ensures that the prescribed radiation exposure conditions are maintained throughout the qualification process.

The radiation dose is generally expressed in **rad(Si)** or **krad(Si)**, representing the amount of energy absorbed by silicon.

6.10 Pre-Irradiation Electrical Characterization

Before exposing a semiconductor device to ionizing radiation, a comprehensive electrical characterization is performed to establish the baseline performance of the device. These baseline measurements serve as a reference for evaluating any degradation caused by radiation exposure. According to **MIL-STD-883 Test Method 1019.9** [3], electrical characterization before irradiation is an essential part of the Total Ionizing Dose (TID) qualification procedure.

The **TLVIH103 High-Speed Comparator Radiation Test Report** [2] follows this methodology by recording the electrical characteristics of the comparator before irradiation. These measurements ensure that the device satisfies all specified electrical requirements prior to radiation exposure and provide reference values for comparison with post-irradiation results.

The parameters generally evaluated during pre-irradiation characterization include:

- Input Offset Voltage
- Positive and Negative Trip Voltages
- Input Bias Current
- Supply Current
- Propagation Delay

- Rise Time
- Fall Time
- Output Logic Levels
- Functional Operation

Each parameter is measured under specified operating conditions using precision laboratory instruments. The measured values are documented and compared with the manufacturer's specifications to verify normal device operation before irradiation.

The baseline data obtained during this stage forms the foundation for assessing radiation-induced degradation after completion of the irradiation process.

6.11 Irradiation Procedure

Following pre-irradiation characterization, the device is subjected to controlled gamma irradiation using the procedure specified in **MIL-STD-883 Test Method 1019.9, Condition A**.

The **TLVIH103 Radiation Test Report** [2] describes the implementation of this procedure using a **Biased High Dose Rate (HDR)** irradiation methodology. During irradiation, the comparator remains under the prescribed electrical bias conditions while being exposed to a calibrated gamma radiation source. Maintaining bias during irradiation allows the test to more closely represent the actual operating conditions encountered in aerospace and defence applications.

The irradiation process generally consists of the following steps:

1. Mount the Device Under Test (DUT) in the irradiation fixture.
2. Apply the specified supply voltage and bias conditions.
3. Position calibrated dosimeters near the DUT to monitor the absorbed radiation dose.
4. Expose the device to the gamma radiation source until the required dose is achieved.
5. Monitor irradiation time and dose rate throughout the experiment.
6. Remove the device after completion of the prescribed radiation exposure.

7. Perform post-irradiation electrical characterization.

The radiation dose is increased in a controlled manner according to the qualification requirements specified in the applicable standard. This controlled exposure ensures repeatability and allows accurate assessment of radiation-induced degradation.

6.12 Post-Irradiation Electrical Characterization

After completion of irradiation, the comparator is subjected to another comprehensive electrical characterization. The purpose of this stage is to determine whether radiation exposure has altered the electrical performance of the device.

The post-irradiation characterization follows the same measurement procedure used during pre-irradiation testing. Maintaining identical measurement conditions ensures that any observed differences are due to radiation effects rather than variations in the experimental setup.

According to the **TLVIH103 Radiation Test Report** [2], the electrical parameters measured after irradiation are compared with their corresponding pre-irradiation values to evaluate the effect of accumulated ionizing dose.

The comparison enables engineers to determine:

- Parameter drift
- Performance degradation
- Functional integrity
- Compliance with electrical specifications
- Radiation tolerance of the device

If the measured parameters remain within the specified limits after irradiation, the device is considered to have successfully met the radiation qualification requirements for the tested dose level.

6.13 Electrical Parameters Evaluated During Radiation Testing

The electrical characterization performed before and after irradiation focuses on the parameters most likely to be influenced by ionizing radiation. The **TLVIH103 High-Speed Comparator Radiation Test Report** evaluates these parameters to determine the radiation tolerance of the device.

7.13.1 Input Offset Voltage

Input Offset Voltage is one of the most sensitive parameters affected by Total Ionizing Dose. Radiation-generated oxide charges and interface traps may shift the input-referred offset voltage, thereby affecting comparator accuracy.

A significant increase in input offset voltage may reduce the precision of threshold detection and degrade the performance of analog signal processing systems.

7.13.2 Supply Current

Radiation exposure may increase the leakage currents within MOS transistors, resulting in higher supply current consumption. Monitoring the supply current before and after irradiation provides information regarding the integrity of the internal circuitry and helps identify abnormal degradation.

7.13.3 Propagation Delay

Propagation delay is a critical dynamic parameter for high-speed comparators. Radiation-induced changes in transistor characteristics may increase the internal switching delay, thereby reducing the operating speed of the comparator.

The propagation delay measured after irradiation is compared with the pre-irradiation value to determine the impact of Total Ionizing Dose on the comparator's dynamic performance.

7.13.4 Output Logic Levels

The output voltage corresponding to the HIGH and LOW logic states is also verified after irradiation. Stable output logic levels indicate that the output driver continues to operate correctly despite radiation exposure.

7.13.5 Functional Verification

In addition to individual electrical parameters, the overall functionality of the comparator is verified after irradiation. Functional testing confirms that the comparator continues to perform its intended operation under the specified electrical conditions.

Successful functional verification indicates that the device remains operational after exposure to the prescribed radiation dose.

6.14 Observations from the Studied Radiation Qualification Report

During this internship, the **TLVIH103 High-Speed Comparator Radiation Test Report** was studied to understand the industrial methodology used for radiation qualification of high-speed comparators.

The study highlighted several important aspects of radiation testing:

- Radiation qualification follows standardized procedures to ensure repeatability and consistency.
- Electrical characterization is performed before and after irradiation using identical measurement conditions.
- Maintaining specified electrical bias during irradiation provides a more realistic assessment of device performance under actual operating conditions.
- Critical electrical parameters such as input offset voltage, propagation delay, and supply current are monitored to evaluate radiation-induced degradation.
- Radiation qualification combines controlled irradiation with detailed electrical characterization to assess device reliability.

The study provided valuable insight into the systematic approach followed by the semiconductor industry for qualifying integrated circuits intended for aerospace and defence applications.

6.15 Industrial Relevance of Radiation Qualification

Radiation qualification plays a vital role in ensuring the reliability and long-term performance of semiconductor devices intended for harsh operating environments. Industries such as aerospace, defence, nuclear power, and high-energy physics require

electronic systems that can continue to function accurately even after prolonged exposure to ionizing radiation. Consequently, radiation testing has become an indispensable part of the semiconductor qualification process.

High-speed comparators are extensively used in satellite communication systems, radar receivers, guidance and navigation electronics, telemetry units, data acquisition systems, and high-speed Analog-to-Digital Converters (ADCs). Since these applications demand high accuracy, low propagation delay, and reliable switching characteristics, even small radiation-induced variations in electrical parameters can significantly affect overall system performance.

The standardized qualification methodology specified in **MIL-STD-883 Test Method 1019.9** [3] enables semiconductor manufacturers to evaluate device reliability under controlled irradiation conditions. By comparing the electrical characteristics before and after irradiation, engineers can determine whether a device satisfies the performance requirements for space and defence applications.

The **TLVIH103 High-Speed Comparator Radiation Test Report** demonstrates how a systematic radiation qualification methodology can be used to evaluate the tolerance of high-speed comparators against Total Ionizing Dose effects. Such qualification procedures are essential for ensuring dependable operation throughout the intended mission life of radiation-hardened electronic systems.

6.16 Learning Outcomes from the Radiation Testing Study

Although radiation exposure experiments were not performed during this internship, studying the radiation qualification methodology provided valuable understanding of industrial semiconductor testing practices.

The study enabled the following learning outcomes:

- Understanding the importance of radiation qualification for aerospace and defence electronics.
- Familiarity with different types of radiation effects affecting semiconductor devices, including Total Ionizing Dose (TID), Single Event Effects (SEE), and Displacement Damage (DD).
- Understanding the concept and significance of **Biased High Dose Rate (HDR)** testing.

- Familiarity with the procedures specified in **MIL-STD-883 Test Method 1019.9, Condition A** for Total Ionizing Dose testing.
- Understanding the role of **Cobalt-60 gamma irradiation** in radiation qualification.
- Knowledge of the importance of maintaining specified electrical bias conditions during irradiation.
- Understanding the purpose of pre-irradiation and post-irradiation electrical characterization.
- Familiarity with the electrical parameters monitored during radiation qualification, including input offset voltage, propagation delay, supply current, and functional verification.
- Appreciation of the systematic qualification procedures followed by the semiconductor industry for high-reliability integrated circuits.

The study complemented the practical characterization work presented in Chapter 6 and provided an overview of the additional qualification requirements necessary for semiconductor devices intended for radiation environments.

References

- [1] Texas Instruments, *TLV3601, TLV3603, TLV3604 325-MHz High-Speed Comparator Datasheet*.
- [2] Texas Instruments, *TLVIH103 High-Speed Comparator Radiation Test Report*.
- [3] *MIL-STD-883, Test Method Standard: Microcircuits, Method 1019.9, Total Ionizing Dose (TID), Condition A*.